

~~Find CFL~~

Construct CFG for following

- 1) $(0+1)^* 000 (0+1)^*$
- 2) $L = \{a^n \mid n \geq 1\}$
- 3) $a^* \cdot b^*$
- 4) $a^+ \cdot b^+$
- 5) $L = \{a^n b^n \mid n \geq 1\}$
- 6) $(ab + (ba)^*)^*$
- 7) $(ab)^+ + (ba)^+$
- 8) $(ab)^* + (ba)^*$
- 9) $L = \{a^n b^{2n+1} \mid n \geq 1\}$
- 10) $L(G) = \{a^n b^i c^j \mid n \geq 1, i \geq 1, j \geq 1\}$
- 11) $L(G) = \{0^i 1^n 2^n \mid i \geq 1, n \geq 0\}$
- 12) $L(G) = \{a^n b a^m \mid n, m \geq 1\}$
- 13) $L(G) = \{a^{2^n} \cdot b^{2^n} \mid n \geq 1\}$
- 14) $L(G) = \{ \underbrace{a^{m+3}}_{\downarrow} b^m \mid m \geq 1 \}$

Note: - treat like $\rightarrow a^3 \cdot a^m \cdot b^m \mid m \geq 1$

- 15) $\{a^n \underbrace{b^m c^m}_{\downarrow} d^n \mid m \geq 0, n \geq 0\}$

Note: - These practise problems for CFG Construction.